

Parameters.

- H_p : prediction horizon (number of time steps)
- H_c : number of controls returned for application, $1 \leq H_c \leq H_p$
- S : number of segments
- T, ℓ : time-step duration (h), segment length (km)
- λ_s : number of lanes in segment s
- d_t, ρ_t^{ds} : prescribed demand (veh/h), downstream density (veh/km/lane)
- u_L, T_h : VSL lower bound (km/h), hold length (positive integer steps)
- $\Delta_{\text{time}}, \Delta_{\text{space}}$: optional temporal and spatial VSL limits (km/h)
- $v_{t,s}^{\text{f}}$: free-flow speed (km/h) (max: $\bar{v}^{\text{f}}$)
- $a_{t,s}$: fundamental-diagram exponent
- $\rho_{t,s}^{\text{crit}}$: critical density (veh/km/lane)
- $q_{t,s}^{\text{cap}}$: per-lane capacity (veh/h/lane)
- $K_{t,s}, \tau_{t,s}$: density offset (veh/km/lane), relaxation time (h)
- $\eta_{t,s}^{\text{high}}$: anticipation coefficient (km^2/h)
- $r_{t,s}, \beta_{t,s}$: on-ramp inflow (veh/h), off-ramp split fraction
- ρ_{max} : merge-model jam density, fixed at 180 veh/km/lane
- $\underline{u}_s$: VSL lower bound (km/h)
- control : control zone (set of segments)

Demand is required at $0 \leq t \leq H_p$ and downstream density at $0 \leq t < H_p$. All $0 \leq s < S$.

Smooth extremum.

$$\text{smin}(x, y) = \frac{x + y - \sqrt{(x - y)^2 + 10^{-4}}}{2}, \quad \text{smax}(x, y) = \frac{x + y + \sqrt{(x - y)^2 + 10^{-4}}}{2}$$

and iterate if there are more than two arguments.

Model variables.

Variable	Meaning	Bounds	Indices
$\text{VSL}_{t,s}$	speed limit (km/h)	$[\underline{u}_s, 150]$	$0 \leq t < H_p$
$\rho_{t,s}$	density (veh/km/lane)	$[10^{-4}, 400]$	$0 \leq t \leq H_p$
$v_{t,s}$	velocity (km/h)	$[10^{-6}, \bar{v}^{\text{f}} + 30]$	$0 \leq t \leq H_p$
Q_t	origin queue (veh)	$[0, 10^4]$	$0 \leq t \leq H_p$
Q_t^{o}	flow admitted from the queue (veh/h)	$[0, q_{0,0}^{\text{cap}} \lambda_0]$	$0 \leq t \leq H_p$
$V_{t,s}^{\text{fd}}$	desired fundamental-diagram speed (km/h)	$[0, \bar{v}^{\text{f}} + 30]$	$0 \leq t < H_p$

Initial conditions.

$$Q_0 = Q^{\text{init}}, \quad Q_0^o = Q^{o,\text{init}}, \quad \rho_{0,s} = \rho_s^{\text{init}}, \quad v_{0,s} = v_s^{\text{init}}.$$

VSL hold length.

$$\text{VSL}_{kT_h,s} = \text{VSL}_{kT_h+1,s} = \dots = \text{VSL}_{(k+1)T_h-1,s} \quad (s \in \text{control}, k \in \mathbb{N})$$

VSL grouping. If `control_changepoints` supplies ordered segment indices $b_0, \dots, b_{M-1}$, set $b_M = S$ and for each $0 \leq i < M$ impose

$$\text{VSL}_{t,b_i} = \text{VSL}_{t,b_i+1} = \dots = \text{VSL}_{t,b_{i+1}-1} \quad (0 \leq t < H_p).$$

Spatial and temporal safety.

$$\begin{aligned} |\text{VSL}_{t,s} - \text{VSL}_{t,s-1}| &\leq \Delta_{\text{space}} & (0 \leq t < H_p, \ s \in \text{control} \cap (\text{control} + 1)). \\ |\text{VSL}_{t,s} - \text{VSL}_{t-1,s}| &\leq \Delta_{\text{time}} & (0 \leq t < H_p, \ s \in \text{control}). \end{aligned}$$

where $\text{VSL}_{-1,\bullet} = \text{VSL}_{\bullet}^{\text{prior}}$ if it exists.

Queue and origin inflow dynamics.

$$\begin{aligned} Q_t &= Q_{t-1} + T (d_{t-1} - Q_{t-1}^o) & (1 \leq t \leq H_p) \\ Q_t^o &= \text{smin} \left(q_{t,0}^{\text{cap}} \lambda_0 \frac{\rho_{\max} - \rho_{t,0}}{\rho_{\max} - \rho_{t,0}^{\text{crit}}}, \quad d_t + \frac{Q_t}{T}, \quad q_{t,0}^{\text{cap}} \lambda_0 \right). \end{aligned}$$

(max of receiving-space limit, available demand, and capacity limit)

Fundamental-diagram dynamics.

$$\begin{aligned} \text{VSL}_{t-1,s} &= v_{t-1,s}^f & (s \notin \text{control}), \\ V_{t-1,s}^{\text{fd}} &= V [\rho_{t-1,s}; v_{t-1,s}^f, a_{t-1,s}, \rho_{t-1,s}^{\text{crit}}] & (s \notin \text{control}), \\ V_{t-1,s}^{\text{fd}} &= \text{smin} (V [\rho_{t-1,s}; v_{t-1,s}^f, a_{t-1,s}, \rho_{t-1,s}^{\text{crit}}], \text{VSL}_{t-1,s}) & (s \in \text{control}). \end{aligned}$$

where

$$V [\rho; v^f, a, \rho^{\text{crit}}] = v^f \exp \left(-\frac{1}{a} \left(\frac{\rho}{\rho^{\text{crit}}} \right)^a \right).$$

Density dynamics.

$$\rho_{t,s} = \rho_{t-1,s} + \frac{T}{\ell \lambda_s} (q_{t-1,s}^{\text{in}} - q_{t-1,s}^{\text{out}}).$$

where inflow and outflow are defined as

$$q_{t-1,s}^{\text{in}} = r_{t-1,s} + \begin{cases} Q_{t-1}^{\text{o}} & s = 0, \\ \rho_{t-1,s-1} v_{t-1,s-1} \lambda_{s-1} & s > 0, \end{cases}$$

$$q_{t-1,s}^{\text{out}} = \frac{\rho_{t-1,s} v_{t-1,s} \lambda_s}{1 - \beta_{t-1,s}}.$$

Velocity dynamics.

$$v_{t,s} = \text{smax}_{\epsilon_v} (\tilde{v}_{t,s}, 10^{-4}).$$

where downstream density, convection, and raw next-step velocity are defined as:

$$\rho_{t-1,s}^{\text{next}} = \begin{cases} \rho_{t-1,s+1} & s < S-1, \\ \rho_{t-1}^{\text{ds}} & s = S-1, \end{cases}$$

$$\text{cv}_{t-1,s} = \mathbf{1}_{s>0} \frac{T v_{t-1,s}}{\ell} (v_{t-1,s-1} - v_{t-1,s})$$

$$\tilde{v}_{t,s} = v_{t-1,s} + \frac{T}{\tau_{t-1,s}} (V_{t-1,s}^{\text{fd}} - v_{t-1,s}) + \text{cv}_{t-1,s} - \frac{\eta_{t-1,s}^{\text{high}} T}{\tau_{t-1,s} \ell} \frac{\rho_{t-1,s}^{\text{next}} - \rho_{t-1,s}}{\rho_{t-1,s} + K_{t-1,s}}.$$

Objective.

$$\min \quad T \sum_{t=1}^{H_p} \left(Q_t + \sum_{s=0}^{S-1} \rho_{t,s} \lambda_s \ell \right).$$